Exploring Neural Columns for Real-Time Information Processing

Technical report

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1 Project idea

Our brain is built of about 10^{11} interacting neural cells that form a large and complex neural network. Different authors estimate a broad range of 75–125 billion neurons in the whole brain [7]. In biological systems, the neurons are organised into neural microcircuits to be able to process complex input information. The function of each microcircuit or column is dependent on neuron types and its location in the brain [3]. The spiking neural networks built of Hodgkin-Huxley (HH) neurons behave like Liquid State Machines (LSM) [6].

We propose a simple, modular HH spiking neural network model called *Ret-Net(5x8,1)*. The model was designed, built and simulated in 2020. We presented our initial results during the 23rd HPI Future SOC Lab Day on November 23rd, 2021. Since then, we performed additional simulations of neural columns, and gathered more data describing their dynamics.

The main scientific objective for this project is to build and simulate LSMs (ranging from 50k to 1M neurons), and to measure their properties.

1.1 Liquid State Machines

The creator of liquid computing theory Wolfgang Maass highlighted that each neuronal column acts as an independent, real-time computation unit called *Liquid State Machine* [8]. Each LSM is capable of processing a continuous stream of multi-modal input from a rapidly changing environment. As he describes, the framework [9] is not only compatible with biological constraints (e.g. his parameters of neurons and synapses were chosen to fit data from microcircuits in rat somatosensory cortex), but it actually requires these biologically realistic features of neural computation to operate better than Touring machines. The abstract model he proposed was inspired by our brains, where heterogeneous columns of neurons are not task-specific [10]. This means that they may possess potentially universal computational capabilities.

The Liquid State Machine simulated in a computer system is therefore a simplified model of the cerebral cortex [12] in which individual neurons are arranged in hyper-columns. These columns, similar anatomically and physiologically to each other, can process several complex tasks in parallel. The general architecture of LSM is presented in Fig. 1. As per definition each Liquid State Machine is built of the three main modules: an *input layer*, at least one *liquid column* or *reservoir*, and the *readout layer*.

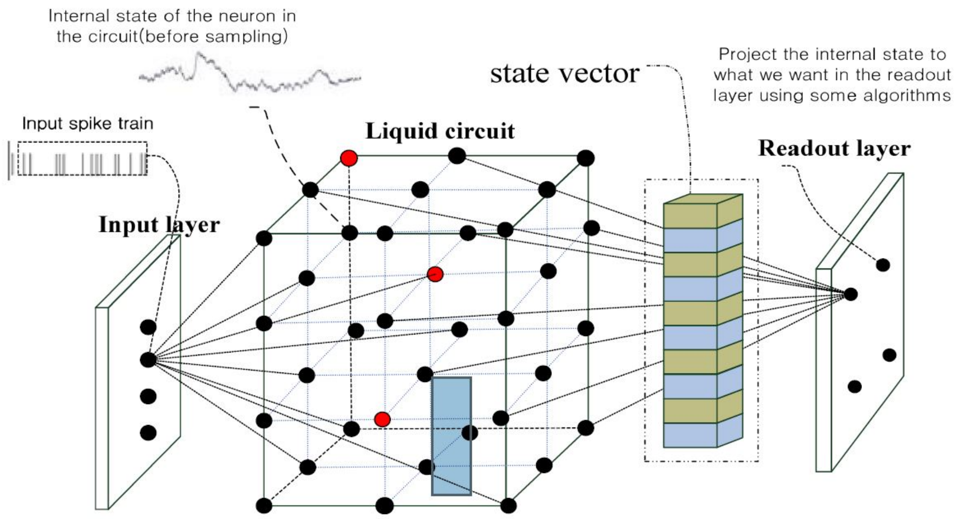

Figure 1: Diagram of Liquid State Machine as presented by Huang [5].

1.2 Objectives

The main objective is to build and examine a large, modular artificial neural network of spiking neurons in a simulation of a LSM system processing visual signals (from 50k to 1M of neurons) on a large computational cluster. The secondary goal is to explore the properties of LSM.

2 Project Activities

2.1 Modelling Visual System

We propose a modular, bio-inspired model of a visual system that consists of two main components, an *Input* (acting as a retina of the system) and *Liquid* (acting as a visual cortex, built of a single LSM column). Our new Hodgkin-Huxley Liquid State Machine (HHLSM) model uses high fidelity multi-compartmental neurons with voltage-activated sodium and potassium channels. We build on a well known conductance-based model describing how action potentials in neurons are initiated and propagated in electrical circuit [13]. The simulation parameters we used can be organised into four groups:

- Main resistances $R_x = 0.3\ \Omega$, $R_n = 0.33\ \Omega$.
- Capacitance $C_n = 0.01$ F and potential incl. $E_n = 0.07$ V, $E_k = 0.0594$ V (soma compartment), $E_k = 0.07$ V (dendrite).
- Conductance $G_K = 360\ \Omega^{-1}$ and $G_{Na} = 1200\ \Omega^{-1}$ (for each of the ionic channels).
- Physical characteristics, a soma with circular shape and diameter of 30 μ, dendrites and axon length of 100 μ.

The exact values for these parameters were achieved experimentally by [13] to provide a high biological fidelity of the model. A detailed description of the Hodgkin-Huxley model can be found in [4].

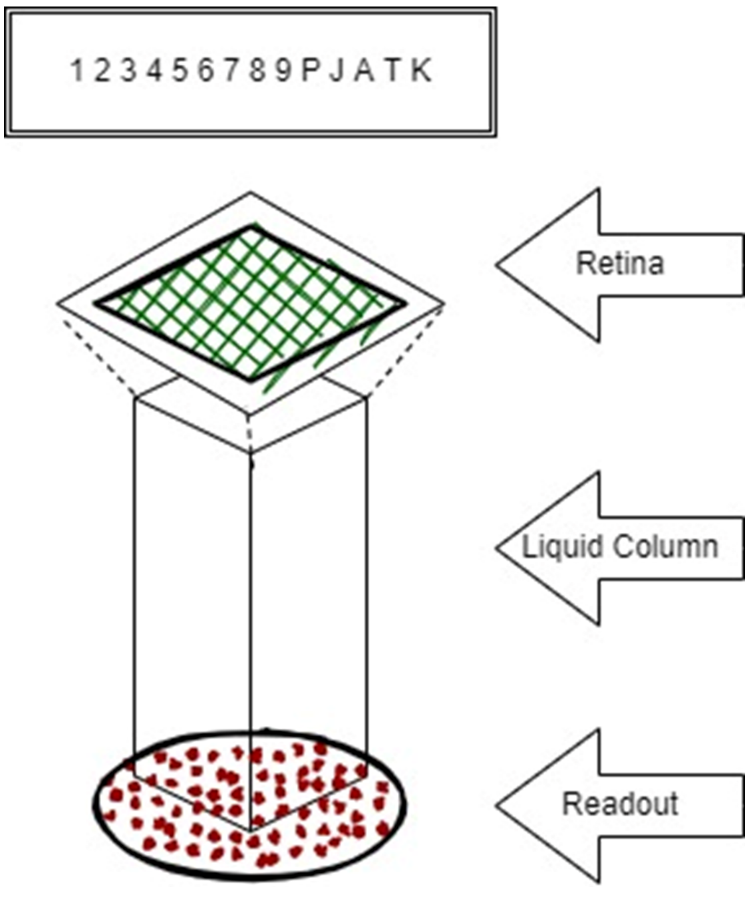

Figure 2: Retina and liquid column of the proposed visual (Input is 15 different patterns).

This model is different to prior models e.g. the one presented in by Wójcik [11] in 2006. The current model called RetNet(5x8, 1) has a single LSM column. It is intended to become a building block of a much larger, and more complex model that could be built of 1M+ neurons, using a novel simulation setup. The final Retina of that target model was built of smaller RetNet(5x8, 1) models on a 125×16 rectangle-shaped grid and divided into 5 patches (25×16), with each patch connected to 10 HHLSM columns forming Lateral Geniculate Nuclei (LGN) and the ensemble of cortical microcircuit. The retinal cells are only connected to the LSM column through the retina's top LGN layer. Each HHLSM was build in 4 versions:

1. 1040 neural cells placed in a rectangular cuboid of 8×5×25.
2. 2040 neural cells placed in a rectangular cuboid of 8×5×50.
3. 3040 neural cells placed in a rectangular cuboid of 8×5×75.
4. 4040 neural cells placed in a rectangular cuboid of 8×5×100.

The preliminary research focuses on Type 1. The structure of each column in the models is the same. As the column can have 4 sizes, assuming 15 stimulating patterns, it is possible to evaluate a LSM system built of progressively 15600, 30600, 45600 or 60600 neurons.

2.2 Future SOC Lab Resources

The main computing resource that we received from Future SOC Lab and used in the project was a powerful Ubuntu 18.04.5 bionic virtual machine. It was used as our development workstation (vm-20211005-001.fsoc.hpi.uni-potsdam.de, running Linux 4.15.0-143-generic on x86_64 architecture). We received a privileged access to the virtual machine allowing us to install the software we needed to compile the latest version of GENESIS simulation environment [1] (version 2.4 from May 2019). The machine was equipped with 8 CPUs: 8 x 2.00 GHz, memory (RAM): 7.79 GB, local disk: we used 7% of 69.38 GB as of April 12th, 2022.

The other resource that we received access to was a shared instance of SAP HANA database. [2]. Similarly to our situation last year, the main reason why we were unable to fully utilise SAP HANA was lack of time to learn how (or design a new solution) to automatically load the files generated by GENESIS simulator to the database. Moreover, as turned out during the simulations our spiking model generated relatively small amounts of data, and it did not require an immediate in-memory processing. Nevertheless, the real-time readout processing with SAP HANA ML services would still be a very interesting topic for further investigation in future, if the access to SAP HANA was extended for the rest of 2022.

2.3 Limitations

Similarly to the 2021 report we state that with the HPI Future SOC Lab's support the key limitation related to lack of computational resources was resolved. When the assigned VM was not available due to the migration of HPI server infrastructure to the new building (e.g. January 15 until January 28, 2022), the authors were running a smaller scale development environment on their Raspberry Pi 4B computer.

3 Preliminary Results

We built and ran 15 RetNet(8x5,1) blocks consisting of 15 LSM columns and 15600 biologically realistic, spiking HH neurons.

We show that our LSM model react differently to 15 different input patterns that are numbers (0-9) and letters (P, J, A, T, K). We performed 225 simulations using the type 1 model. We measured CPU Time and Memory consumption for the models, as well as evaluated the simulation setup. The aggregated results of the simulations are presented in Fig 6, Fig 7, Fig 8.

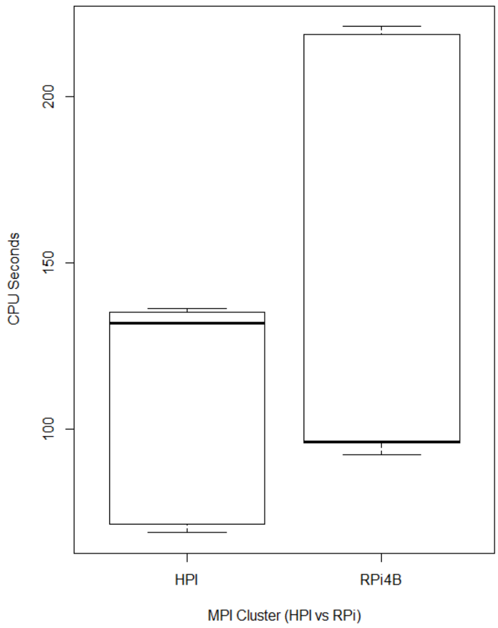

Figure 3: The box plot presenting simulation time for different clusters.

4 Next Steps

The modular HH spiking neural network model RetNet(5x8,1) was run . The main objective was to build a large model, and to measure its separation property. We

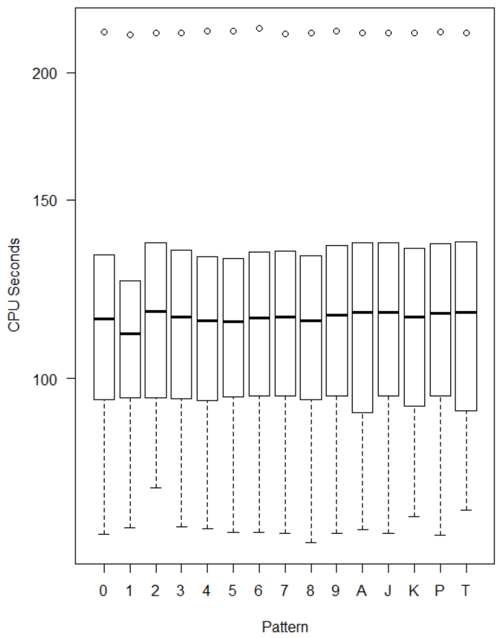

Figure 4: The box plot presenting simulation time for different input patterns.

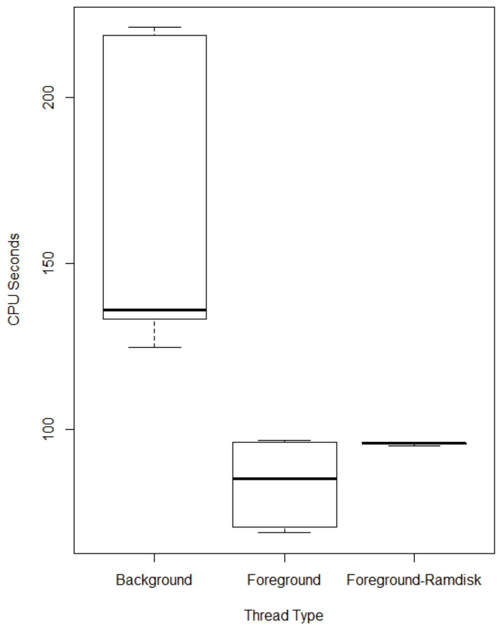

Figure 5: The box plot presenting simulation time for different input patterns.

have progressed towards the main goals. Until November 2021 we only simulated 2k HH neurons organised in a two part rectangle-shaped grid (retina) and 2 cortical microcircuits (1 node, 8 CPU cores) using HPI resources. By April 2022 we simulated the system built of 15 columns and 60600 neurons.

In the upcoming months we would like to continue our research to run larger models (e.g. 1M+ model), and focus on incorporating Slurm and/or Docker into our pipeline. We hope that HPI Future SOC Lab will allow us to experiment with Slurm Workload Manager and building multi-architecture Docker containers for GENESIS.

5 Acknowledgements

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6 Appendix

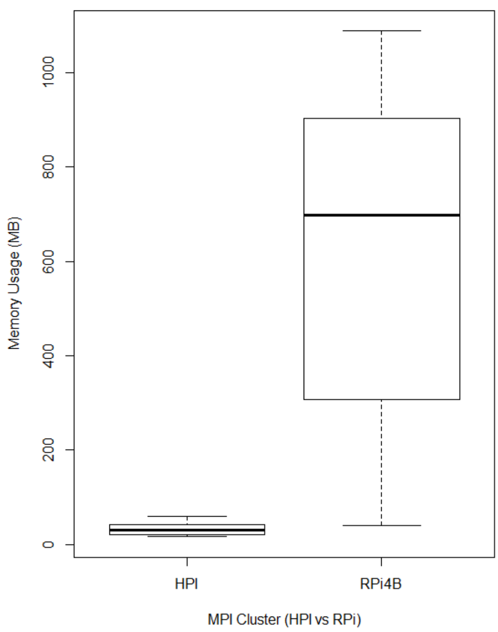

Figure 6: The box plot presenting memory utilisation for different clusters.

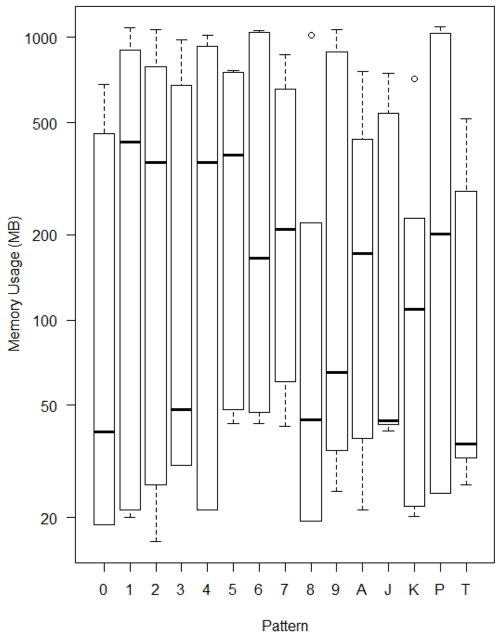

Figure 7: The box plot presenting memory utilisation for different input patterns.

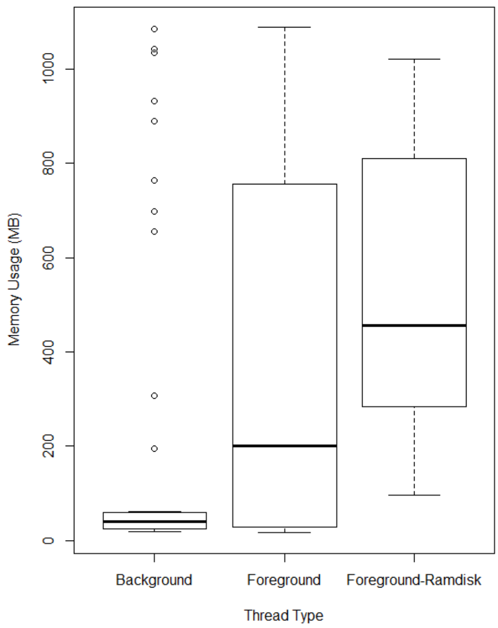

Figure 8: The box plot presenting memory utilisation for different input patterns.

References

[1] J. M. Bower, D. Beeman, and M. Hucka. "The GENESIS simulation system". In: (2003).

[2] F. Färber, N. May, W. Lehner, P. Große, I. Müller, H. Rauhe, and J. Dees. "The SAP HANA Database–An Architecture Overview." In: *IEEE Data Eng. Bull.* 35.1 (2012), pages 28–33.

[3] A. Gupta, Y. Wang, and H. Markram. "Organizing principles for a diversity of GABAergic interneurons and synapses in the neocortex". In: *Science* 287.5451 (2000), pages 273–278.

[4] A. L. Hodgkin and A. F. Huxley. "A quantitative description of membrane current and its application to conduction and excitation in nerve". In: *The Journal of physiology* 117.4 (1952), pages 500–544.

[5] J. Huang, Y. Wang, and J. Huang. "The Separation Property Enhancement of Liquid State Machine by Particle Swarm Optimization". In: *International Symposium on Neural Networks*. Springer. 2009, pages 67–76.

[6] W. A. Kamiński and G. M. Wójcik. "Liquid state machine built of Hodgkin-Huxley neurons-pattern recognition and informational entropy". In: *Annales Universitatis Mariae Curie-Sklodowska, sectio AI–Informatica* 1.1 (2015), pages 1–7.

[7] R. Lent, F. A. Azevedo, C. H. Andrade-Moraes, and A. V. Pinto. "How many neurons do you have? Some dogmas of quantitative neuroscience under revision". In: *European Journal of Neuroscience* 35.1 (2012), pages 1–9.

[8] W. Maass, T. Natschläger, and H. Markram. "A model for real-time computation in generic neural microcircuits". In: *Advances in neural information processing systems* 15 (2002).

[9] W. Maass, T. Natschläger, and H. Markram. "Real-time computing without stable states: A new framework for neural computation based on perturbations". In: *Neural computation* 14.11 (2002), pages 2531–2560.

[10] G. M. Shepherd. *A basic circuit for cortical organization*. MIT-Press, 1988.

[11] G. M. Wojcik and W. A. Kaminski. "Liquid computations and large simulations of the mammalian visual cortex". In: *International Conference on Computational Science*. Springer. 2006, pages 94–101.

[12] T. Yamazaki and S. Tanaka. "The cerebellum as a liquid state machine". In: *Neural Networks* 20.3 (2007), pages 290–297.

[13] T. Yorozu, M. Hirano, K. Oka, and Y. Tagawa. "Electron spectroscopy studies on magneto-optical media and plastic substrate interface". In: *IEEE translation journal on magnetics in Japan* 2.8 (1987), pages 740–741.